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WIRESHARK MANUAL

1. Purpose of Wireshark

Wireshark is a free and open-source **network protocol analyzer** (also known as a packet sniffer or network analyzer). It captures live network traffic from various interfaces and allows interactive browsing and deep inspection of packet data.

Key purposes include:

- **Network troubleshooting**: Identify connectivity issues, latency, retransmissions, or errors.
- **Protocol analysis and development**: Inspect how protocols behave "on the wire," debug implementations, or reverse-engineer communications.
- **Security analysis**: Detect anomalies, malware communication (e.g., C2 via DNS), intrusions, or policy violations.
- **Education and learning**: Understand network protocols through real packet captures.
- **Forensics**: Analyze saved captures (.pcap/.pcapng files) for post-incident review.

Wireshark provides detailed dissection of hundreds of protocols, live capture from many network media, offline analysis, and powerful display/capture filters. It is cross-platform (Windows, macOS, Linux/Unix) and relies on libraries like libpcap/WinPcap/Npcap for capturing.

2. How to Approach a Packet Capture

A systematic approach ensures efficient and accurate analysis:

1. **Define the objective**: What problem are you solving? (e.g., slow website loading → focus on HTTP/DNS/TCP; connectivity failure → ICMP/TCP handshake).
2. **Select the capture location**: Capture as close as possible to the issue (client-side, server-side, or mirrored port on a switch). Avoid capturing on the same busy machine if possible.

3. **Choose the interface**: In Wireshark's welcome screen, select the correct network interface (Ethernet, Wi-Fi, etc.). Use ***ipconfig/ifconfig/ip*** link to identify it.
4. **Apply capture filters:** Use BPF syntax to reduce noise and file size (e.g***., host 8.8.8.8, port 53, udp port 53, tcp port 80 or tcp port 443***). Avoid over-filtering if unsure.
5. **Start the capture**: Double-click the interface or use Capture → Start (or Ctrl+E). Perform the test/action that reproduces the issue.
6. **Stop and save**: Stop capture (red square or Capture → Stop), then save as .pcapng (preferred format).
7. **Apply display filters**: In the filter bar, use Wireshark's display filter language (e.g., ***dns, http, tcp.analysis.retransmission, ip.src == 192.168.1.10***).
8. **Analyze hierarchically**:
    - Packet List pane: Overview (time, source/dest, protocol, info).
    - Packet Details pane: Expand protocol layers (frame → eth → ip → tcp/udp → dns/http/etc.).
    - Packet Bytes pane: Raw hex/ASCII view.
    - Use Follow → TCP/UDP Stream or HTTP Stream for conversation reconstruction.
9. **Use tools & statistics**: Conversations, Endpoints, Flow Graph, I/O Graph, Expert Information (for errors/warnings), Resolved Addresses.
10. **Iterate**: Apply color rules, export objects (e.g., HTTP files), or create custom columns.

3. Protocol Deep Dive: DNS (Domain Name System)

DNS is the **phonebook of the Internet**. Without it, we would have to remember IP addresses (like 192.168.1.1 or 2400:cb00:2048:1::c629:d7a2) instead of typing simple names like google.com or nytimes.com.

What is DNS?

- DNS stands for **Domain Name System**.
- It translates human-readable **domain names** (example.com) into machine-readable **IP addresses** (142.250.190.78).
- Every device on the internet has a unique IP address. DNS removes the need for humans to memorize these long numbers.
- It works completely “behind the scenes” — you never see it happening when you open a website.

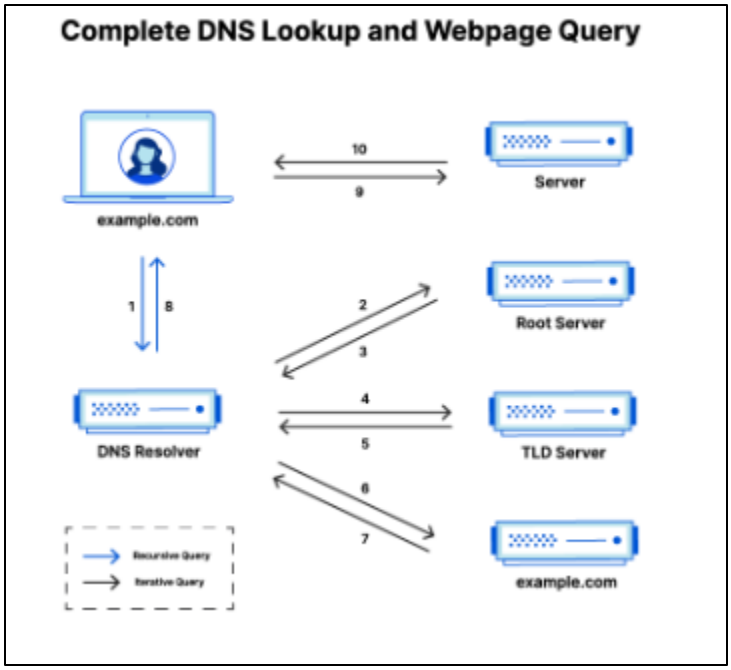
How Does DNS Work?

When you type a website name in your browser, a DNS **lookup** (resolution) happens. The browser needs an IP address to connect. DNS does this translation through a chain of special servers.

4 DNS Servers Involved in Loading a Webpage:

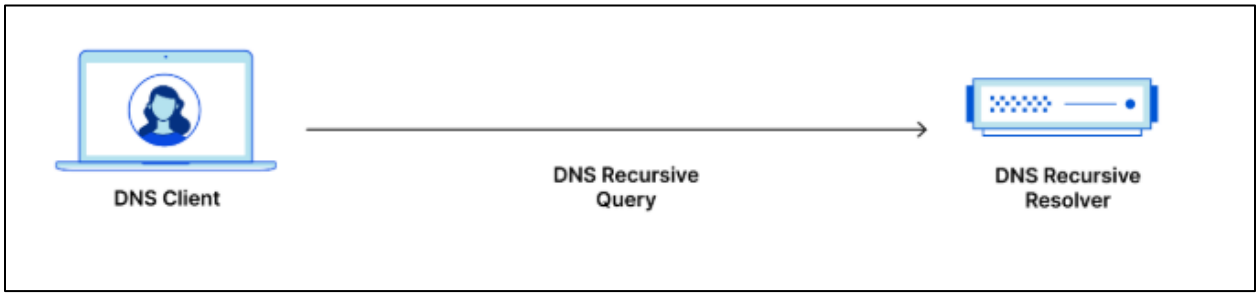
There are exactly **4 types of servers** that work together:

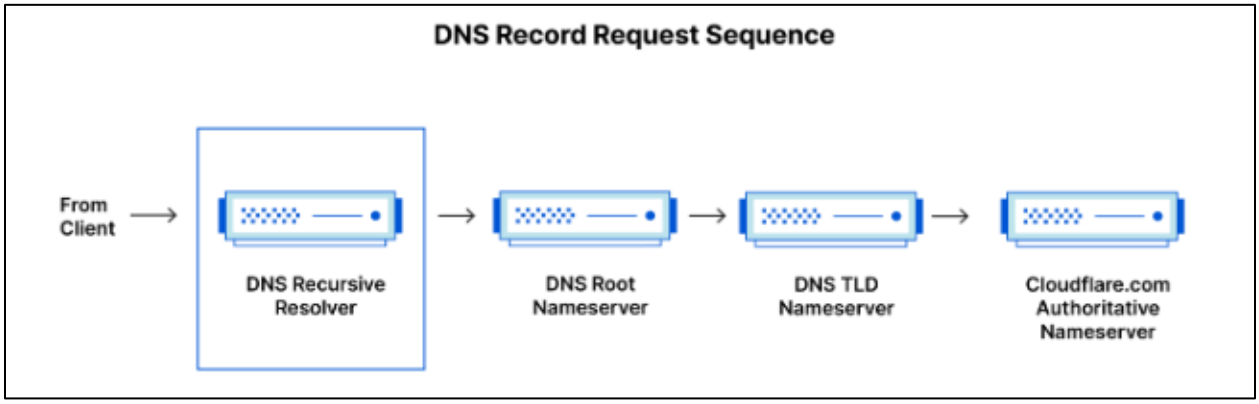
Server Type	Role (Simple Analogy)	What it does
DNS Recursor	Librarian	Takes your request and does all the hard work to find the answer
Root Nameserver	Library Index	Points to the correct TLD (like .com, .org, .pk)
TLD Nameserver	Specific Rack of Books	Knows which authoritative server has the domain
Authoritative Nameserver	Actual Dictionary on the rack	Gives the final IP address of the website



Recursive Resolver vs Authoritative Nameserver (Most Important Difference)

- **Recursive Resolver** → Starts the query (your ISP or 8.8.8.8). It keeps asking other servers until it gets the answer.
- **Authoritative Nameserver** → The final owner of the domain (e.g., Google's own server for google.com). It gives the real IP address.





8 Steps in a DNS Lookup (When Nothing is Cached)

This is the **full uncached process** (takes only milliseconds):

1. You type example.com in browser → request goes to **Recursive Resolver**.
2. Resolver asks **Root Nameserver** (.).
3. Root replies: "Ask the .com TLD server".
4. Resolver asks **.com TLD Nameserver**.
5. TLD replies: "Ask the authoritative server for example.com".
6. Resolver asks **Authoritative Nameserver**.
7. Authoritative server replies with the **IP address**.
8. Resolver sends the IP back to your browser → website loads.

Note: If any step is already cached, many of these steps are skipped → super fast.

3 Types of DNS Queries

1. **Recursive Query** — "Give me the full answer or error" (client to resolver).
2. **Iterative Query** — "Give me the best answer you have or next server" (resolver to root/TLD).
3. **Non-Recursive Query** — "I already know you have the record" (fastest, when cached).

DNS Caching –Why Everything is Fast

DNS records are temporarily stored (cached) so the same query doesn't have to travel the whole world again.

Caching happens at 4 places:

- **Browser cache** (first check)
- **Operating System (OS) cache** (stub resolver)
- **Recursive Resolver** (ISP or Google DNS)
- **Authoritative server** itself (with TTL value)

TTL = Time To Live → How long the record should stay in cache (e.g., 300 seconds = 5 minutes).

DNS in Wireshark (Practical Analysis Tips)

- Filter: dns (all DNS) or dns.flags.response == 0 (only queries)
- Look for:
    - dns.qry.name → queried domain
    - dns.flags.response → 0 = query, 1 = response
    - dns.qry.type → 1 = A (IPv4), 28 = AAAA (IPv6)
    - Source IP = your machine
    - Destination IP = DNS server (usually 8.8.8.8 or your router)
- Right-click → Follow → UDP Stream to see full conversation.

Reference: https://www.cloudflare.com/learning/dns/what-is-dns/